

MS4215: Lab 7

Robust Regression, Ridge, Lasso, and Cross-Validation: Baseball Salaries

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Introduction

The dataset `hitters.csv` contains 1986 MLB season statistics for 263 baseball players with non-missing salary data. The variables are:

Variable	Description
Player	Player name
AtBat	Number of times at bat in 1986
Hits	Number of hits in 1986
HmRun	Number of home runs in 1986
Runs	Number of runs in 1986
RBI	Number of runs batted in during 1986
Walks	Number of walks in 1986
Years	Number of years in the major leagues
CAtBat / CHits / CHmRun / CRuns / CRBI / CWalks	Career totals
League	Player's league at end of 1986 (A or N)
Division	Player's division at end of 1986 (E or W)
PutOuts / Assists / Errors	Fielding statistics
Salary	1987 annual salary (\$000s) — outcome
NewLeague	Player's league at start of 1987

The lab covers three topics from the lecture. **Part A** fits an ordinary least squares (OLS) model, checks for influential outliers, and compares to robust regression. **Part B** applies ridge and lasso regularisation. **Part C** estimates predictive performance using K-fold cross-validation.

Before attempting the questions, run the setup chunk below to load the required packages and data.

```
#install.packages("tidyverse")
#install.packages("MASS")
#install.packages("glmnet")
#install.packages("caret")
library(tidyverse)
library(MASS)
library(glmnet)
library(caret)

# In the line of code below, you will need to change the location to where you
# have saved the file
df <- read_csv("data/hitters.csv")
df$log_Salary <- log(df$Salary)
```

```
df$League    <- factor(df$League)
df$Division  <- factor(df$Division)
df$NewLeague <- factor(df$NewLeague)
```

Questions

Part A: OLS and robust regression

(i) Data exploration

Plot histograms of `Salary` and `log_Salary` side by side. Use `summary()` to inspect the full dataset. Comment on the shape of the salary distribution and why a log transformation is appropriate.

(ii) OLS baseline and outlier diagnostics

Fit an OLS regression of `log_Salary` on five predictors:

```
m_ols <- lm(log_Salary ~ Years + Hits + Walks + RBI + HmRun, data = df)
```

Report the coefficient table and adjusted R^2 . Then plot `plot(m_ols, which = 5)` (residuals vs leverage). Which players have the highest Cook's distances? Do any clearly stand out?

(iii) Robust regression

Fit the same model using `rlm()` from the MASS package:

```
m_rlm <- rlm(log_Salary ~ Years + Hits + Walks + RBI + HmRun, data = df)
```

Compare the coefficients and scale/RSE from both models. Extract the Huber weights using `m_rlm$w` and inspect the three players with the lowest weights (pipe directly from `df` using `mutate()` rather than saving the weights back into `df`). Conclude: does OLS appear to be distorted by outliers here, or does `rlm()` confirm the OLS results?

Part B: Ridge and lasso regularisation

(iv) Ridge regression

Build the full predictor matrix using `model.matrix()` (19 predictors, with factors dummy-encoded), and the response vector:

```
x_mat <- model.matrix(log_Salary ~ . - Salary - Player, data = df)[, -1]
y_vec <- df$log_Salary
```

Use `set.seed(1)` then `cv.glmnet(x_mat, y_vec, alpha = 0)` to select the best lambda by cross-validation. Fit the ridge model at `lambda.min` and display the coefficients. Are all 19 predictors retained?

(v) Lasso regression

Use `set.seed(2)` then `cv.glmnet(x_mat, y_vec, alpha = 1)` to fit lasso. Display the coefficients. How many predictors does lasso set to exactly zero? Which predictors survive (non-zero coefficients)? What does this selection tell us about which variables drive salary?

Part C: K-fold cross-validation

(vi) Estimated predictive performance

Use `caret::train()` to evaluate the 5-predictor OLS model via 10-fold cross-validation (repeated 3 times):

```
set.seed(125)
train_control <- trainControl(method = "repeatedcv", number = 10, repeats = 3)
cv_model <- train(log_Salary ~ Years + Hits + Walks + RBI + HmRun,
```

```
data = df, method = "lm", trControl = train_control)
```

Report the CV RMSE and CV R^2 . Compare the CV R^2 with the training R^2 from `m_ols`. Is there evidence of overfitting? What does this tell us about using a 5-predictor model on 263 observations?